

COURSE NOTES

Matrix Computation

Instructor:

Prof. SHANGZHI ZENG

Student:

Leo

February 26, 2026

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Chapter 1

Lecture 1

Course Information

This course establishes the mathematical foundation for matrix analysis and numerical computations, serving as a cornerstone for modern applications in machine learning, artificial intelligence, computer vision, and data science.

The core curriculum addresses two fundamental classes of problems:

1. Solving Linear Systems ($Ax = b$):

- *Square Systems* ($A \in \mathbb{R}^{n \times n}$): Solved via direct methods.
- *Overdetermined Systems* ($A \in \mathbb{R}^{m \times n}, m > n$): Addressed using least squares or minimum norm solutions.
- *Underdetermined Systems* ($m < n$): Explored through sparse solutions.

2. Eigenvalue Problems ($Av = \lambda v$):

- We investigate numerical algorithms for spectral analysis, noting that algebraic solutions for characteristic polynomials of degree ≥ 5 are generally unavailable.

A critical distinction is drawn between **direct methods** and **iterative methods**. While direct methods provide exact solutions, their computational cost can be prohibitive for large-scale matrices. Consequently, we emphasize efficient iterative techniques—such as the Conjugate Gradient method and Krylov subspace methods (Lanczos, GMRES)—which are pivotal for high-dimensional computation.

• Course Scope:

- Basic matrix concepts, subspaces, and norms.
- Linear least squares and pseudo-inverses.
- Eigenvalue and Singular Value Decompositions (EVD, SVD); positive semidefinite matrices.
- Linear system solvers: LU, Cholesky, QR, and Schur decompositions.
- Krylov subspace methods (Lanczos, GMRES, etc.).